Anthony Schmidt, Ph.D

Data Science and Applied Research

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acircleda

Curriculum Vitae
August 2025

Professional Appointments

2021-present	**Senior Data Scientist** Amplify Education	Remote/Brooklyn, NY
2019-2021	**Data Visualization Researcher** College and University Professional Association for Human Resources (CUPA-HR)	Knoxville, TN
2014-2019	**Teaching Associate** English Language Institute, University of Tennessee	Knoxville, TN
2012-2014	**English Language Instructor** Department of English Language and Literature, Pukyong National University	Busan, South Korea

Education

2022 **PhD, Educational Psychology and Research** — Knoxville, TN
University of Tennessee, Knoxville

- Concentration: Evaluation, Statistics, and Methodology
- Minor: Statistics and Data Science
- Dissertation: Science Education to the Rescue? Assessing the Relationship Between Scientific Literacy and Carbon Emissions

2011 **MSEd, Language, Literacy, and Cultural Education** — Bloomington, IN
Indiana University, Bloomington

- Concentration: TESOL

2006 **BA, Anthropology** — Tampa, FL
University of South Florida

Professional Research Experience

2024-2025 **Opportunity Scoring**
Amplify Education

- Optimized ML opportunity scoring model through experimentation
- Translated model insights into actionable guidance for sales teams.
- Built Looker pipelines for monitoring and score transparency

2025 **Standards Matcher**
Amplify Education

- Built a semantic similarity prototype tool with sentence transformers to automate standards crosswalks

2025 **HS Math Pilot Analyses**
Amplify Education

- Analyzed usage and performance data to guide content authors in refining item difficulty and development

2021-Present **Internal Tooling**
Amplify Education

- Developed and maintained internal R/Python packages to streamline data access, ML, and reporting.

2021-2025 **Embedded Assessment**
Amplify Education

- Productionized an R-based EM service, generating 200K+ student scores daily.
- Collaborated on a EM data and models to deliver real-time math insights to teachers.

2023 **Teacher Comment Analysis**
Amplify Education
- Analyzed teacher comments using descriptive stats, n-grams, sentiment, and topic models to inform feedback tool development.

2022, 2024 **Hackathon Projects**
Amplify Education
- 2022: Built R Shiny dashboard visualizing single-user journeys across the platform.
- 2024: Evaluated ML, BERT, and LLM models for spam detection; built Streamlit POC with Snowflake Cortex.

2021-2023 **Closing the Loop**
Amplify Education
- Automated quarterly math item quality report with psychometric and ML measures (difficulty, discrimination, readability, isolation forest).
- Improved item quality and influenced authoring practices based on report insights.

2021 **Minority-Serving Institutions**
CUPA-HR
- Analyzed salary and diversity trends at MSIs by integrating internal CUPA-HR data with IPEDS, IPUMS, and other sources.

Graduate Research Experience

Software

Schmidt, A., et al. (2025). *pyAmp* [internal Python package].
Janda, K., Gamble, H., Lama, T., Crane, S., **Schmidt, A.** (2023). *amplifyR* [internal R package].
Rocconi, L. & **Schmidt, A.** (2022). *mlmhelpr* [R package]. https://cran.r-project.org/web/packages/mlmhelpr/index.html
Schmidt, A.& Kulma, K. (2020). *footprint* [R package]. https://cran.r-project.org/web/packages/footprint/index.html

Publications

Peer Reviewed Journal Articles

Schmidt, A. (2022). *University air travel and greenhouse gas mitigation: an analysis of higher education climate policies*. International Journal of Sustainability in Higher Education. *23*(6). https://doi.org/10.1108/IJSHE-07-2021-0318

Schmidt, A. (2020). *Are international students getting a bang for their buck? The relationship between expenditures and international student graduation rates*. Journal of International Students. *10*(3). https://www.ojed.org/index.php/jis/article/view/1279. https://files.eric.ed.gov/fulltext/EJ1265702.pdf

Schmidt, A. (2016). *Listening journals for extensive and intensive listening practice*. English Teaching Forum. *54*. https://eric.ed.gov/?id=EJ1104020

Pre-Prints and Unpublished Manuscripts

Schmidt, A. & Rocconi, L. (2021). *STARS and stripes: Political and institutional factors that affect higher education sustainability ratings*. EdArXiv. 10.35542/osf.io/d5bku. https://doi.org/10.35542/osf.io/d5bku

Schmidt, A. (2021). *Comparing intensive English program (IEP) and directly-admitted international students' academic achievement*. EdArXiv. 10.35542/osf.io/vm37t. https://doi.org/10.35542/osf.io/vm37t

Book Reviews

Schmidt, A. (2021). *Review of the book International students in first-year writing: A journey through socio-academic space, by M. Siczek*. Journal of International Students. *9*(3). https://doi.org/10.32674/jis.v0i0.907

Technical Reports

Schmidt, A. (2022, February). *The representation of women and racial/ethnic minorities in the workforce of minority-serving higher education institutions* [Research report]. CUPA-HR. https://www.cupahr.org/surveys/research-briefs/women-minorities-msi-workforce-february-2022/

Schmidt, A., & Jones, S. N. (2021, February). *Sustainabe energy attitudes report* [Research report]. University of Tennessee, Knoxville. https://acircleda.shinyapps.io/SustainableEnergySurvey/

Pritchard, A. & **Schmidt, A.** (2020, July). *The Higher Ed HR Workforce* [Research report]. CUPA-HR. https://www.cupahr.org/surveys/research-briefs/

Bichsel, J., McChesney, J., & **Schmidt, A.** (2019). *Focus on psychology faculty: Salaries, pay equity, minority representation, and the growth of non-tenure-track faculty* [Research report]. CUPA-HR. https://www.cupahr.org/surveys/research-briefs/

Nadel, S., **Schmidt, A.**, & Bichesel, J. (2019). *The impact of the FSLA overtime rule on higher education* [Research report]. CUPA-HR. https://www.cupahr.org/surveys/research-briefs/

Schmidt, A., Rocconi, L., Morrow, J. A., Berry, B. (2019). *College of Architecture and Design's undergraduate advising 2019 report* [Research report]. Evaluation, Statistics, and Measurement (ESM) Program Research Group.

Morrow, J. A., Rocconi, L., Butler, B., & **Schmidt, A.** (2019). *University of Tennessee's undergraduate advising 2019 evaluation end of year report* [Research report]. Evaluation, Statistics, and Measurement (ESM) Program Research Group.

Conference Papers and Presentations

International/National Conferences

Schmidt, A. (2021, October). *University air travel and GHG mitigation: An analysis of higher education climate policies* [Paper presentation]. Global Conference on Sustainability in Higher Education.

Schmidt, A. & Rocconi, L. (2021, August). *From click to code: Insights from instructors and students on moving from SPSS to R in a graduate-level applied statistics course* [Paper presentation]. Joint Statistical Meetings.

Schmidt, A. (2020). *Principles and practices for great data visualization* [Conference session]. CUPA-HR Annual Conference.

Rosenberg, J., **Schmidt, A.**, Rosenberg, A., Longnecker, J., & Mann, M. (2020, June). *Becoming tidier over time: Studying #tidytuesday as a social media-based context for learning to visualize data* [Paper presentation]. International Society of the Learning Sciences. https://repository.isls.org/handle/1/6455

Schmidt, A. (2019, October). *A comparison of direct admit and IEP student academic achievement* [Paper presentation]. TESOL International Conference, Atlanta, Georgia.

Schmidt, A. (2017, March). *Off course: A comparison of university and EAP coursebook writing tasks* [Paper presentation]. TESOL International Conference, Seattle, Washington..

Regional Conferences

Rosenberg, J. M., Hodge, L., Aydeniz, M., **Schmidt, A.**, Lishinski, A., Rich, K., Longnecker, J., Mann. M., & Sadovnik, A. (2020, January). *A survey of teachers and administrators regarding the implementation of new K-8 computing education standards in Tennessee* [Paper presentation]. 14th Annual Tennessee STEM Education Research conference, Cookeville, Tennessee.

Schmidt, A. (2016, October). *Academic reading circles* [Conference session]. Southeast TESOL Regional Conference, Louisville, Kentucky.

Schmidt, A. (2016, May). *Principled washback: Integrating test prep to foster academic skills* [Conference session]. Toronto TESOL Conference, Toronto, Canada.

Schmidt, A. (2016, February). *Academic reading circles and collaborative meaning construction* [Conference session]. GATESOL IEP Mini-Conference, Atlanta, Georgia.

Schmidt, A. (2015, October). *Audio diaries for improved spoken grammatical accuracy and pronunciation* [Conference session]. Southeast TESOL Regional Conference, New Orleans, Louisiana.

Poster Presentations

Schmidt, A. (2020, March). *Becoming "tidyr" over time: Data visualization development in an online community of practice* [Poster presentation]. University of Tennessee, Knoxville College of Education, Health, and Human Sciences Graduate Research Colloquium.

Schmidt, A. (2019, March). *A comparison of direct-admit and IEP undergraduate international student achievement* [Poster presentation]. University of Tennessee, Knoxville College of Education, Health, and Human Sciences Graduate Research Colloquium.

Schmidt, A. & Chitty, E. (2018, May). *Making history: Supporting language through content in intensive English programs* [Poster presentation]. NAFSA Annual Conference, Philadelphia, Pennsylvania.

Campus & Department Talks

Schmidt, A. (2020, October). *University air travel and GHG mitigation: Examining institutional policies through the lens of #flyingless* [Invited presentation]. Committee for the Campus Environment, University of Tennessee, Knoxville.

Schmidt, A. & Chitty, E. (2018, March). *Teaching international students: Improving learning for all* [Workshop]. Teaching and Learning Innovation, University of Tennessee, Knoxville..

Schmidt, A. (2016, November). *Intensive reading for learner engagement* [Conference session]. Re-Imagine: Teaching and Learning Innovation Symposium, University of Tennessee, Knoxville..

Invited Talks

Schmidt, A. (2018, March). *Two hot topics in OLL research: Corpus linguistics and telecollaboration* [Webinar]. OLLReN, Cambridge University Press.

Professional Workshops, Webinars, and Trainings

Schmidt, A. (2025, August). *SQL + Python: Integrated analysis in VSCode* [Workshop]. Amplify Education.

Schmidt, A. (2025, March). *Python package development* [Workshop]. Amplify Education.

Schmidt, A. (2024, March). *Data learning series part 2: Visualizing* [Workshop]. Amplify Education.

Schmidt, A. (2020, December). *Research skills lab: Data visualization basics* [Workshop]. CUPA-HR.

Schmidt, A. (2020, October). *Data visualization in R* [Workshop]. Fall 2020 Evaluation, Statistics, and Measurement seminar, University of Tennessee, Knoxville..

Schmidt, A. (2020, April). *Tableau training: Building your first dashboard* [Workshop]. Spring 2020 Evaluation, Statistics, and Measurement seminar, University of Tennessee, Knoxville..

Teaching Experience

Full-Time Experience

2014-2019 **English Language Institute, University of Tennessee** Knoxville, TN
Teaching Associate
- Courses (non-credit): Grammar, Reading, Writing, Listening/Speaking
- Levels: Beginner to Advanced

2012-2014 **Department of English Language and Literature, Pukyong National University** Busan, South Korea
English Language Instructor
- Undergraduate (3-credit): Intermediate Composition, Advanced Composition, Lower-Intermediate Conversation, Upper-Intermediate Conversation

Other Teaching Experience

2023 Spring **University of Tennessee, Knoxville** Knoxville, TN
Clinical Instructor
- Designed syllabus and course materials to teach Data Visualization in R as part of the Educational Data Science Graduate Certificate

2021 Spring **University of Tennessee, Knoxville** Knoxville, TN
Graduate Teaching Assistant
- Designed materials to teach R for Applied Statistics III

2018 (Summer) **Southeast University** Nanjing, China
Visiting Instructor, Department of Foreign Languages
- Course: Cross-Cultural Communication (1 credit, undergraduate)

2012-2014 **Samsung Electronics/Pukyong National University** Busan, South Korea
English Language Instructor
- Course: English Communication (3 credits, Intercultural Communication Undergraduate Program)

2010-2012 **Yangun Middle School** Busan, South Korea
English as a Foreign Language Instructor

2009 **Tsukuno Junior High School** Osaka, Japan
English as a Foreign Language Instructor

2007-2009 **Daicheong Middle School** Busan, South Korea
English as a Foreign Language Instructor

Teacher Training Experience

2012 **Pukyong National University** Busan, South Korea
In-Service Secondary School Teacher Trainer
- National English Ability Test (NEAT) Writing Composition Training Program

2010-2012 **Busan Education Training Institute** Busan, South Korea
In-Service Primary School Teacher Trainer

Awards and Honors

2022 **Hackathon Winner: "A Day in the Life" Shiny App**
Amplify Education

2020 **Friends of Educational Psychology and Counseling Research Grant** $400
University of Tennessee

2020 **Graduate Student Colloquium Travel Award for Best Poster** $200
University of Tennessee

2019 **Graduate Student Senate Travel Award** $780
University of Tennessee

2018 **Diversity Professional Development Fund for Staff** $1,000
University of Tennessee

2018 **Travel Fund** $500
Tennessee Association of International Educators

2017 **Travel Grant for IEP Instructors** $1,500
University of Pittsburgh

2012 **Teaching Excellence Award** $300
Pukyong National University, Department of Humanities

Skills

Programming Languages	R, Python, SQL, JavaScript, HTML, CSS
DevOps	Git, AWS, Docker, Terraform, dbt
Reporting	Quarto, Markdown, Shiny, Looker, Tableau, Streamlit
Statistics	Machine Learning, Predictive Modeling, Statistical Inference, Psychometrics, Experimental Design, Time Series
Other Software	SPSS, Fusion360, TinkerCAD
Natural Languages	English (native), French (intermediate), Korean (basic)

Notes

This CV is programmatically generated with R. All source code is available on Github.